Problem Set #27

UMass Recreational Math Club

February 2026

Problem 1. How many trailing zeros are there in 1000! (1000 factorial)?

Problem 2. If you want to cover all the vertices in a 5×5 square grid by lines, none of which is parallel to a side of the square, how many lines are needed?

Bonus: generalize to an $n \times n$ grid.

Problem 3. Before you is a circular saw and a $3 \times 3 \times 3$ wooden cube that you must cut into twenty-seven $1 \times 1 \times 1$ cubelets. What's the smallest number of slices you must make in order to do this? You are allowed to stack pieces prior to running them through the saw.

Problem 4. Alice and Bob play a game where at each step one of them puts a stone in one of the squares of the 8×8 board. A player loses if they create an L-shape of three stones. What is a winning strategy?

Problem 5. Let S be any set of 20 distinct integers chosen from the arithmetic progression $1, 4, 7, \cdots, 100$. Prove that there must be two distinct integers in S whose sum is 104.